

Minimum Degree Conditions for Cycles Including Specified Sets of Vertices

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Abstract

This paper generalises the concept of vertex pancyclic graphs. We define a graph as set-pancyclic if for every set S of vertices there is a cycle of every possible length containing S . We show that if the minimum degree of a graph exceeds half its order then the graph is set-pancyclic. We define a graph as k -ordered-pancyclic if, for every set S of cardinality k and every cyclic ordering of S , there is for every possible length a cycle of that length containing S and encountering S in the specified order. We determine the best possible minimum-degree condition which guarantees that a graph is k -ordered-pancyclic.

1 Introduction

All our graphs are simple and undirected. A graph is *pancyclic* if there is a cycle of every length from 3 up to the order. This property was introduced by Bondy [1], who showed that minimum degree greater than half the order ensures the property. Much has been written about generalisations of this condition, by for example, replacing the minimum degree condition with degree conditions on sets of vertices.

Bondy [2] also defined that a graph is *vertex pancyclic* if there is for each vertex a cycle of every length containing that vertex. He showed that the same minimum degree is sufficient:

Theorem 1 [2] *Let G be a graph with order n and minimum degree δ . If $\delta \geq (n+1)/2$, then G is vertex pancyclic.*

He also proved a sufficient lower bound for graphs to be edge pancyclic. Williamson [9] generalised edge pancyclic graphs and showed that if $\delta \geq n/2 + 1$, then the graph is *panconnected*; that is, for every pair u, v of distinct vertices there is a $u-v$ path of every length from 2 up to $n-1$.

In this paper we consider two generalisations of Theorem 1.

1.1 Set-pancyclic graphs

The first generalisation considers the property where for every set of vertices there is a cycle of every possible length containing that set. Whether or not a length is considered possible is determined only by the subgraph $\langle S \rangle$ induced by the set S . In particular, the **path cover number** $pc(H)$ of a graph H is the minimum number of vertex-disjoint paths that use up all the vertices of H . To simplify the statement of the results, it is useful to define that a graph with a hamiltonian cycle has path cover number 0. So, the shortest possible cycle through S has length $|S| + pc(\langle S \rangle)$.

So, for $k \geq 2$, we say that a graph is ***k*-vertex-pancyclic** if every set S of k vertices is in a cycle of every length from $k + pc(\langle S \rangle)$ up to n . We say that a graph is ***set-pancyclic*** if this property holds for all $k \geq 2$.

There are a few points to note. First, one can extend the definition to $k = 1$ to get the original property of vertex pancyclic, though one must define allowable cycles lengths as 3 up to n . Further, 2-vertex-pancyclic implies vertex pancyclic.

Second, for large k , specifically $k > n/2$, this definition is somewhat peculiar in that it does not guarantee any cycle at all through the k vertices. For example, the empty graph on n vertices is k -vertex-pancyclic for $k > n/2$. (However, for the range of minimum degrees given in the next theorem, the graph is guaranteed to be hamiltonian and this phenomenon does not occur.)

We show the following:

Theorem 2 *Let G be a graph with order n and minimum degree δ . If $\delta \geq (n + 1)/2$, then G is set-pancyclic.*

The condition $\delta \geq (n + 1)/2$ cannot be improved. Even if one considers only a fixed k , in general the complete bipartite graph $K(n/2, n/2)$ shows that $\delta \geq (n + 1)/2$ is best possible. (However, the values $k = n$ and $k = n - 1$ are exceptions: a graph is always n -vertex-pancyclic, and for $k = n - 1$, $\delta \geq n/2$ suffices.)

1.2 k-ordered-pancyclic graphs

The second generalisation is where the order of encountering the vertices in the set is specified. This is motivated by the definition by Ng and Schultz [8] of

***k*-ordered-hamiltonian** graphs; that is, graphs where for every k vertices there is a hamiltonian cycle which encounters them in any specified order.

Ng and Schultz considered what minimum degree forces the property of k -ordered-hamiltonicity and gave a partial result. Further results are given by Kierstead et al. [7] and Faudree et al. [5], but the actual threshold is not known for all combinations of n and k . The results of Faudree et al. [5] suggest that the threshold for k -ordered-hamiltonian is exactly the same as the threshold for k -ordered. (A graph is ***k*-ordered** if for every k vertices there is a cycle which encounters them in any specified order. See [4].)

For $k \geq 2$, we define a graph to be ***k*-ordered-pancyclic** if every set S of k vertices are in a cycle of every possible length with specified order. By possible, we look at the “ordered path cover number” $opc(\langle S \rangle)$ of S ; this equals the number of edges omitted between pairs of consecutive vertices in S . (If $k = 2$, then we define $opc(\langle S \rangle)$ as 1 if the two vertices of S are adjacent, and 2 otherwise.) So, the required cycle lengths are from $|S| + opc(\langle S \rangle)$ up to n .

We show that the sufficient minimum-degree condition for k -ordered-pancyclic is higher than that for k -ordered-hamiltonian for $k \leq n/2$. (For $k > n/2$ the reverse holds; this is possible because the condition k -ordered-pancyclic does not imply k -ordered-hamiltonian for that range.)

Theorem 3 *Let G be a graph with order n and minimum degree δ , and let $k \geq 2$ be an integer. For G to be k -ordered-pancyclic it is sufficient that:*

- (a) $\delta \geq (n + 1)/2$ for $k \in \{2, 3\}$;
- (b) $\delta \geq (n + 3)/2$ for $k \in \{4, 5\}$;
- (c) $\delta \geq n/2 + k - 3$ for $6 \leq k \leq (n + 5)/3$ and k even;
- (d) $\delta \geq n/2 + k - 7/2$ for $7 \leq k \leq (n + 5)/3$ and k odd;
- (e) $\delta \geq (3n + k - 9)/4$ for $(n + 6)/3 \leq k \leq n - 4$;
- (f) $\delta \geq n - 2$ for $n - 3 \leq k \leq n - 1$.

And these results are best possible.

Part (a) is immediate from the previous theorem, since k -vertex-pancyclic and k -ordered-pancyclic are equivalent for $k \in \{2, 3\}$.

For both theorems, the approach in the proof is to prove the existence of the smallest cycle and then prove that one can repeatedly extend the cycle.

The proof of Theorem 3 uses considerable numerical calculations (linear programming and brute-force search). A Mathematica program with these calculations is available at <http://www.cs.clemson.edu/~goddard/papers/orderedpan>.

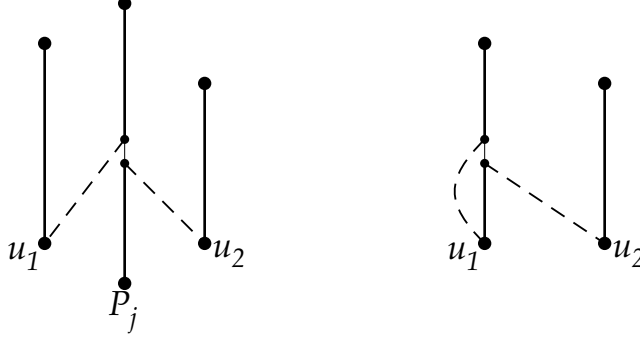


Figure 1: Forbidden pairs of edges

2 Proof of Theorem 2

Let A and B be sets of vertices. If A and B are disjoint, then the **degree-sum** of A in B is the number of edges with one end in A and one end in B . Similarly, if $A \subseteq B$ then the **degree-sum** of A in/inside B is the number of edges with both ends in B and at least one end in A .

Lemma 1 *Let G be a graph of order n with minimum degree $\delta \geq (n+1)/2$. If S is a subset of the vertices, $|S| \geq 2$, with path cover number m , then there exists a cycle of length $|S| + m$ through S .*

PROOF. If $m = 0$ then we are done as the graph $\langle S \rangle$ is hamiltonian. So assume $m \geq 1$. Consider a minimum path-cover $\mathcal{P} = \{P_1, \dots, P_m\}$ of $\langle S \rangle$. For $1 \leq i \leq m$, let u_i and v_i denote the two endpoints of the path P_i if P_i has length at least 1, otherwise let $u_i = v_i$ be the vertex of P_i .

Assume first that $m \geq 2$. Note that by the minimality of $\mathcal{P}$, if two vertices of the collection $\{u_1, \dots, u_m, v_1, \dots, v_m\}$ are adjacent, then they must be endpoints of the same path. Consider u_1 and u_2 . For $j \geq 3$, by minimality, there cannot be two consecutive vertices on P_j such that one is a neighbour of u_1 and one a neighbour of u_2 . See Figure 1. Since neither u_1 nor u_2 is adjacent to an endpoint of P_j , that means that the degree-sum of $\{u_1, u_2\}$ in P_j is at most $|P_j| - 1$ (actually only achievable for $|P_j|$ odd).

Furthermore, by minimality of $\mathcal{P}$, on P_1 there cannot be two consecutive vertices such that a neighbour of u_1 comes after a neighbour of u_2 . So the

degree-sum of $\{u_1, u_2\}$ in P_1 is at most $|P_1| - 1$. Similarly, the degree-sum of $\{u_1, u_2\}$ in P_2 is at most $|P_2| - 1$.

Thus, the degree-sum of $\{u_1, u_2\}$ inside S is at most $|S| - m$. By the minimum degree condition, they have degree-sum outside S (that is, degree-sum in $V(G) - S$) at least $n + 1 + m - |S|$. This means that u_1 and u_2 have at least $m + 1$ common neighbours outside S . This result applies to any two endpoints of different paths of $\mathcal{P}$.

Thus, we can find m distinct vertices $w_1, \dots, w_m$ outside S , with each w_i a common neighbour of u_i and v_{i+1} (arithmetic modulo m). These vertices can be added to the paths P_i to produce the desired cycle.

Finally, suppose $m = 1$. If $|S| = 2$, then u_1 and v_1 have a common neighbour, which yields the desired triangle (this is the only point that one needs $\delta > n/2$). So assume $|S| \geq 3$. If u_1 is adjacent to v_1 , then we have a contradiction (as then $m = 0$). Furthermore, on P_1 there cannot be two consecutive vertices such that a neighbour of u_1 comes after a neighbour of v_1 (as then $m = 0$ again). So the degree-sum of $\{u_1, u_2\}$ inside S is at most $|S| - 1$, and so they have two common neighbours outside S . QED

The remaining cycles are found one a time using the following result of Hendry [6]. For a given cycle, we call a vertex outside the cycle an **external vertex**. We say that a graph is **cycle extendible** if for every nonhamiltonian cycle C there exists an external vertex v such that the graph induced by $V(C) \cup \{v\}$ has a spanning cycle.

Theorem 4 [6] *Let G be a graph with order n and minimum degree $\delta \geq (n+1)/2$. Then G is cycle extendible.*

Hendry [6] actually proved an Ore-type version. It follows from Lemma 1 above and his results that:

Theorem 5 *Let G be a graph of order n satisfying $\deg v + \deg w \geq (3n - 3)/2$ for all nonadjacent vertices v and w . Then G is set pancyclic.*

PROOF. Hendry [6] showed that $\deg v + \deg w \geq (4n - 5)/3$ is sufficient for a graph to be cycle extendible. It is easy to check that the proof of Lemma 1 still goes through if the minimum-degree condition is replaced by $\deg v + \deg w \geq (n+1)/2$, except in the case where $|S| = 2$ and $m = 1$. That is, it remains only to show that the given condition ensures that any two adjacent vertices lie in a triangle.

Let u and v be adjacent vertices, and suppose they have no common neighbour. Let y be a nonneighbour of u and x a nonneighbour of v . Then $\deg u + \deg v \leq n$ and $\deg x, \deg y \leq n - 2$ so that $\deg u + \deg v + \deg x + \deg y \leq 3n - 4$; but by the Ore condition, $\deg u + \deg y + \deg v + \deg x \geq 2(3n - 3)/2$, a contradiction. QED

This result is best possible by taking the disjoint union of K_2 and K_{n-2} and making each vertex of the K_2 adjacent to at least $\lfloor (n - 2)/2 \rfloor$ vertices in the big clique such that each vertex in the big clique is adjacent to at most one vertex in the K_2 . The two vertices of the K_2 are not in a triangle.

3 Proof of Theorem 3

We start with the construction of the graphs which show that the bounds in Theorem 3 are best possible. Then we show that minimum degree at least $n/2 + k - O(1)$ is sufficient for a graph to be k -ordered-pancyclic. After this which we show that for $k > n/3 + O(1)$, minimum degree at least $(3n + k)/4 - O(1)$ is sufficient.

3.1 Constructions

For positive integers a, b and n with $a \geq 4$ and $n \geq a + b$, define a graph $G_n(a, b)$ on n vertices as follows. Let $c = n - a - b$. Start with five cliques A_1, A_2, B, C_1, C_2 of cardinalities $\lfloor a/2 \rfloor, \lceil a/2 \rceil, b, \lfloor c/2 \rfloor$, and $\lceil c/2 \rceil$ respectively. Then add all edges between the cliques, except that there are no edges between A_1 and C_2 , and no edges between A_2 and C_1 . Finally, if a is even then remove the a edges of a cycle that alternates between A_1 and A_2 , and if a is odd then remove the $a - 1$ edges of a path that alternates between A_1 and A_2 starting and finishing with A_2 . The graph $G_n(a, b)$ has minimum degree $\min(n - \lceil c/2 \rceil - 3, n - \lceil a/2 \rceil - 1)$ provided $c > 0$. See Figure 2.

Now, if a is even, then $G_n(a, a - 1)$ is not a -ordered-pancyclic: for, consider the set $A_1 \cup A_2$ with the vertices in the order specified in the missing cycle. Each of the a consecutive pairs needs a common neighbour, but only B is available and B is too small.

Also, if $a \leq b + 2$, then the graph $G_n(a, b)$ is not $(b + 2)$ -ordered-pancyclic: for, consider a set of vertices containing $A_1 \cup A_2$, with those vertices in the order specified in the missing cycle/path, followed by $b + 2 - a$ vertices of B . Then each

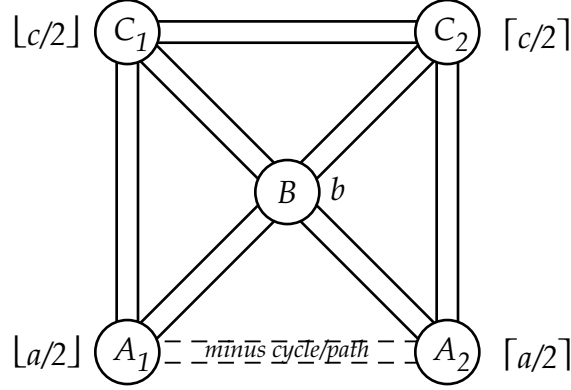


Figure 2: The graph $G_n(a, b)$

of the $a - 1$ consecutive pairs of vertices in $A_1 \cup A_2$ needs a common neighbour, but only B is available and what remains of B is too small.

For the construction with $5 \leq k \leq (n + 5)/3$, proceed as follows. For k even, consider $G_n(k, k - 1)$. This is not k -ordered-pancyclic, and for this range of k the minimum degree is $n - \lceil c/2 \rceil - 3 = \lfloor (n - 7)/2 \rfloor + k$. For k odd, consider $G_n(k, k - 2)$. This is not k -ordered-pancyclic, and for this range of k the minimum degree is $n - \lceil c/2 \rceil - 3 = \lfloor (n - 8)/2 \rfloor + k$.

For the construction with $(n + 6)/3 \leq k \leq n - 4$, proceed as follows. Consider $G_n(\lfloor (n - k + 6)/2 \rfloor, k - 2)$ (so that the degrees of vertices in A_i and C_i are as close as possible). The minimum degree is $\lfloor (3n + k - 10)/4 \rfloor$. For this range of k , it holds that $a \leq b + 2$, and so the graph is not k -ordered-pancyclic.

For the construction with $k = 4$, define a graph H_n as follows. Start with four cliques A_1, A_2, C_1, C_2 of cardinalities 2, 2, $\lfloor n/2 \rfloor - 2$, and $\lceil n/2 \rceil - 2$ respectively. Then add all edges between the cliques, except that there are no edges between A_1 and C_2 , and no edges between A_2 and C_1 . The graph H_n has order n and minimum degree $\lfloor n/2 \rfloor + 1$. It is not 4-ordered-pancyclic, as there is no 5-cycle which takes in $A_1 \cup A_2$ where the vertices of A_1 and A_2 alternate.

For the construction with $k \geq n - 3$, proceed as follows. Consider the graph $G_n(6, n - 6)$ formed by taking the complete graph K_n and removing the edges of a 6-cycle $v_1, v_2, v_3, \dots, v_6, v_1$. The graph has minimum degree $n - 2$. To show that it is not k -ordered-pancyclic for $k = n - 2$ or $n - 3$, take a set S containing v_1, v_2, v_4 and v_5 , but neither v_3 nor v_6 , such that v_1, v_2, v_4, v_5 appear consecutively. Then one needs a joint neighbour of v_1 and v_2 , and of v_4 and v_5 , but at most one

vertex is available. For $k = n - 1$, take S containing all but v_3 and ensure that v_1 and v_2 appear consecutively but no other pair of vertices of the cycle does.

3.2 Sufficiency for k small: Initial cycle(s)

In this and the next subsection we prove that minimum degree $n/2 + k - O(1)$ is sufficient for a graph to be k -ordered. The basic outline is as follows. We first prove the existence of the smallest required cycle through the designated set S ; this uses Hall's matching theorem. Then we consider extending the cycle. If there is an external vertex adjacent to two consecutive vertices on the cycle, then this is trivial. If not, the subgraph induced by the external vertices is dense, and so pan-connected. Then the idea is to find two disjoint edges between the cycle and the external vertices, such that the segment of the cycle between the two edges contains no vertex of S ; thus one can replace the segment of the cycle with paths of all lengths via the external vertices. The problem is that the segment might be too short—so that one or two cycle lengths are missed—or too long—so that we make no progress. These problems can be easily overcome for a slightly larger minimum degree or if k is large, but the exact result when k is small requires many details.

We start with the smallest cycle. We will repeatedly use Hall's or König's matching theorem (see [3]).

Lemma 2 *Let $k \geq 4$ be an integer. Let G be a graph with order n and minimum degree $\delta \geq n/2 + k - 7/2 + \varepsilon_k$ with $\varepsilon_k = \frac{1}{2}$ if k even, and $\varepsilon_k = 0$ if k is odd. Let S be a cyclically ordered subset of k vertices with $\text{opc}(\langle S \rangle) = m$. If $k + m \leq n$, then there is a cycle of length $k + m$ through S in the requisite order.*

PROOF. Define an **open pair** as a pair of consecutive vertices of S which are not adjacent, and define a **closer** for the pair as an external vertex which is adjacent to both members. There are m open pairs.

Let P be an open pair. Suppose there are z_P other open pairs involving a vertex of P (note that $z_P \in \{0, 1, 2\}$). Then the degree-sum of P inside S is at most $2k - 4 - z_P$.

(a) Assume k is even. Then the degree-sum of P outside S is at least

$$(n + 2k - 6) - (2k - 4 - z_P) = (n - k) + (k - 2 + z_P).$$

If $m = k$, then every open pair P has $z_P = 2$, and thus has k closers. Hence there are k distinct closers, one for each pair. If $m = k - 1$, then every open pair has $z_P \geq 1$ and so has at least $k - 1$ closers. If $m \leq k - 2$, then every pair has at least $k - 2$ closers. In any case, we can find the cycle of length $k + m$ by using a distinct closer for each open pair.

(b) Assume k is odd. Then the degree-sum of P outside S is at least

$$(n + 2k - 7) - (2k - 4 - z_P) = (n - k) + (k - 3 + z_P).$$

Assume $m = k$. Then every open pair has $z_P = 2$, and thus has at least $k - 1$ closers. By Hall's theorem we are done, unless every open pair has exactly $k - 1$ closers and these $k - 1$ closers are the same for all k pairs. For an open pair to have exactly $k - 1$ closers, by its degree-sum every other external vertex must be adjacent to exactly one member of the pair. Consider three consecutive vertices u_1, u_2, u_3 of S . Since $\{u_1, u_2\}$ and $\{u_2, u_3\}$ have the same set of closers, and since every external vertex is adjacent to at least one of $\{u_1, u_2\}$ and one of $\{u_2, u_3\}$, it follows that u_1 and u_3 have the same external neighbours. This implies that any pair of vertices of S two apart in the cyclic ordering have the same external neighbours. But since k is odd, this means that all vertices of S have the same external neighbours. This is a contradiction since $k - 1 = m - 1 < n - k$. Hence we can find k distinct closers, one for each open pair.

If $m = k - 1$, then every open pair has $z_P \geq 1$ and so has $k - 2$ closers, and some pair P^* has $z_{P^*} \geq 2$ and so has $k - 1$ closers. If $m \leq k - 2$, then every pair has at least $k - 3$ closers; and if $m = k - 2$, then some pair has $z_{P^*} \geq 1$ and so $k - 2$ closers. In any case, the conditions of Hall's theorem are satisfied, and so there exist distinct closers for each open pair and thus the cycle of length $k + m$ exists. QED

The process of growing the cycle in the next subsection can miss a cycle length or two. So we will need the following lemma.

Lemma 3 *Let $k \geq 4$ be an integer. Let G be a graph with order n and minimum degree $\delta \geq n/2 + k - 7/2 + \varepsilon_k$ with $\varepsilon_k = 1$ if $k = 4$, $\varepsilon_k = \frac{1}{2}$ if $k \geq 6$ and even, and $\varepsilon_k = 0$ if k is odd. Let S be a cyclically ordered subset of k vertices with $\text{opc}(\langle S \rangle) = m$. If $k + m + 2 \leq n$, then there are cycles of length $k + m + 1$ and $k + m + 2$ through S in the requisite order.*

PROOF. We can grow the cycle of length $k + m$ guaranteed by the previous lemma if there is an external vertex that is adjacent to two consecutive vertices

on the cycle. So assume that does not happen twice. That is, we have a cycle C of length $d \in \{k + m, k + m + 1\}$ where no external vertex has two consecutive neighbours on the cycle. Note that

$$d \leq 2k + 1, \quad (1)$$

and by the hypothesis that

$$d \leq n - 1. \quad (2)$$

Since an external vertex is adjacent to at most half the cycle,

$$\delta - (n - d - 1) \leq d/2. \quad (3)$$

We claim that

$$d \geq k + 1. \quad (4)$$

Any vertex on C has at least $\delta - d + 1$ external neighbours. Since two consecutive vertices on C have no common external neighbour, it follows that $2(\delta - d + 1) \leq n - d$ so that $d \geq 2\delta - n + 2 \geq 2k - 5 + 2\varepsilon_k$. This is at least $k + 1$ except when $k = 5$. So suppose $d = k = 5$. Then the external degree-sum of two consecutive vertices on C is exactly $n - d$; hence a pair of vertices two apart on C have the same external neighbours. Since k is odd, this yields a contradiction. Hence we have shown that $d > k$.

There are now two cases.

(a) Every vertex of C has at least one external neighbour.

Consider any two consecutive vertices u and v on C . Then u has an external neighbour, call it u' . Note that the external-degree of v is at least $\delta - d + 1$ and of u' is at least $\delta - d/2$, and that u' and v are not adjacent. Then we claim that u' and v have common external neighbour z . For, otherwise we have the system of inequalities:

$$\delta \geq (n - 7)/2 + \varepsilon_k, (1), (2), (3), (4), \delta \leq n - 1, \text{ and } (\delta - d + 1) + (\delta - d/2) \leq n - d - 1.$$

Using linear programming, one can show that this system has no solution unless $k = 5$ and $d \geq 10$. But in this case $m \geq 4$, and so one can choose v to be a member of S with at least two nonneighbours in S ; thus v has external degree (See Check A1)

at least $\delta - d + 3$, and after a little more calculation the claim follows. Hence one obtains the $(d + 2)$ -cycle by replacing uv by the path $uu'zv$.

Further, consider vertices u and v on C with exactly one vertex between them and with that vertex x not in S (exists since $d > k$). Again let u' be an external neighbour of u and we claim that u' and v have a common external neighbour z . Otherwise we obtain a similar system of inequalities, except that as u' may be adjacent to v , there is the slightly weaker bound $(\delta - d + 1) + (\delta - d/2) \leq n - d$. Again linear programming is used; this shows that the system has no solution unless $k \leq 5$ and $d \geq 8$. (See

In that case, since d is large relative to k , we may choose $v \in S$ to have at least one nonneighbour in S and thus at least $\delta - d + 2$ external neighbours. This revised system of inequalities is the same as the first system above, which had no solution unless $k \leq 5$ and $d \geq 10$. As before, we can then choose u to have two nonneighbours in S , and thus finally show that u' and v have a common external neighbour z . Hence one may obtain the $(d + 1)$ -cycle by replacing the path uxv by the path $uu'zv$. Check A2)

(b) *Some vertex of C has no external neighbour.*

So we have the system of integer inequalities:

$$\delta \geq (n - 7)/2 + \varepsilon_k, (1), (2), (3), (4), \text{ and } \delta \leq d - 1.$$

If one maximises n subject to the linear programming relaxation of this system, one obtains $n \leq 17$. By brute force trying of all possibilities, it then follows that the only integral solutions to the above system are (n, d, k, δ) either $(15, 10, 5, 9)$ or $(17, 11, 5, 10)$. (See

Such examples do occur. But it is not hard to show that in this case, if the cycle is not immediately extendible, then there is the following situation: there are four consecutive vertices u_1, u_2, u_3, u_4 on C , with an external vertex x adjacent to u_1 and u_3 , with u_2 adjacent to u_4 and not both u_2 and u_3 in S . Then to obtain a $(d + 1)$ -cycle one replaces $u_1u_2u_3u_4$ by $u_1xu_3u_2u_4$. QED Check A3)

3.3 Sufficiency for k small: Extending cycles

For the next result we need some terminology. We call a cycle D **problematic** if no external vertex is adjacent to two consecutive vertices on D .

Lemma 4 *Let G be a graph with minimum degree $\delta \geq (n+3)/2$ and let D be a problematic cycle. Then the subgraph $G-D$ is panconnected and D has length at most $2(n-\delta-1)$.*

PROOF. The graph $G-D$ has minimum degree at least $\delta-|D|/2 \geq (n-|D|+3)/2$. Thus the first part follows from Williamson's result. Also, since $\delta-|D|/2 \leq n-|D|-1$, the upper bound on $|D|$ holds. QED

Given a set S of specified vertices on a cycle D , we say that a pair of vertices $\{u, v\}$ on D is **detourable**, if the segment between them on D contains no vertex of S and they have distinct external neighbours. (Note that the vertices u and v themselves may be in S .) We let $t(D)$ denote the shortest length of the segment between a detourable pair, if such a pair exists.

Lemma 5 *Let D be a problematic cycle on $d \leq n-5$ vertices containing a set S of k vertices.*

- (a) *Let T be positive integer with $T \leq 10$. If $t(D) \geq T+1$ or $t(D)$ does not exist, then $Td \leq Tk + (T+1)(n-\delta-1)$.*
- (b) *If there are at least $k+2$ external vertices which have at least $k+2$ neighbours on D , then $t(D) \leq d/2 - k/2$.*

PROOF. (a) Choose an orientation of D and partition D into segments as follows: start a new segment at every vertex of S or when the current segment has $T+1$ vertices, whichever comes first. Let X denote the number of segments. Then $X \leq (k-1) + \lceil (d-(k-1))/(T+1) \rceil \leq (Tk+d)/(T+1)$.

Since $t(D) \geq T+1$, for each segment there is either only one vertex on that segment with external neighbours, or only one external vertex with neighbours on that segment. Thus there are at most $\max(n-d, \lceil T/2 \rceil)$ edges from that segment to exterior vertices. By assumption, $\max(n-d, \lceil T/2 \rceil) = n-d$.

Every external vertex has at least $\delta - (n-d-1)$ neighbours on the cycle. So, if M is the number of edges between external vertices and the cycle, then

$$(n-d)(\delta - (n-d-1)) \leq M \leq X(n-d) \leq ((Tk+d)/(T+1))(n-d).$$

This simplifies to the desired inequality.

(b) Again partition D into segments. This time, start at each vertex of S ; so there are k segments. The second-longest segment has at most $(d-(k-2))/2$ vertices. The longest segment has at most $d-(k-1)$ vertices; divide it into two

segments each with at most $\lceil (d - k + 1)/2 \rceil$ vertices. So now every segment has at most $(d - k)/2 + 1$ vertices.

Since there are $k + 1$ segments, and there are $k + 2$ exterior vertices with at least $k + 2$ neighbours on D , by the pigeon-hole principle, there must be a segment such that two exterior vertices both have two neighbours on that segment. Thus that segment contains a detourable pair. So $t(D)$ exists and $t(D) \leq (d - k)/2$. QED

Given a set S of specified vertices on a cycle D , we define a *splice* vertex as a vertex on the cycle that is not in S and whose predecessor and successor on the cycle are adjacent.

We are now in a position to prove the main result for k small.

Lemma 6 *Let $k \geq 4$ be an integer. Let G be a graph with order n and minimum degree $\delta \geq n/2 + k - 7/2 + \varepsilon_k$ with $\varepsilon_k = 1$ if $k = 4$, $\varepsilon_k = \frac{1}{2}$ if $k \geq 6$ and even, and $\varepsilon_k = 0$ if k is odd. Then G is k -ordered-pancyclic.*

PROOF. Let S be a cyclically ordered set of k vertices. By Lemma 2, there is the requisite smallest cycle C_1 through S . We grow the cycle by repeatedly inserting an external vertex adjacent to two consecutives on the cycle until a hamiltonian cycle is reached, and are thus done, or a problematic cycle is encountered.

The minimum degree conditions for $k = 4$ and $k = 5$ are the same. Consider $k = 4$. By Lemmas 2 and 3, the smallest and second-smallest cycle are obtained. The second-smallest cycle has length at least 5; thus the remaining cycles are implied by the result for $k = 5$. That is, it suffices to prove the result for $k \geq 5$.

Say the first problematic cycle is D_1 of length d_1 .

Case 1: $d_1 \leq \delta$.

Then, every vertex on D_1 has at least one external neighbour. So, since D_1 is problematic, any two consecutive vertices u and v have distinct external neighbours u' and v' respectively.

Since $G - D_1$ is panconnected (by Lemma 4), u' and v' have paths of all possible lengths between them in $G - D_1$. The requisite cycles of all lengths from $d_1 + 3$ up to n follow by replacing the edge uv in D_1 with a path via u' and v' . It remains to find the cycles of length $d_1 + 1$ and $d_1 + 2$.

Every time we grow the cycle, a vertex is inserted such that its predecessor and successor on the new cycle are adjacent. That is, the new vertex is a splice

vertex. So, if there are two splice vertices x_1 and x_2 in D_1 , choose u and v to be both distinct from x_1 and x_2 . Then one can remove first x_1 , then x_2 , from the cycle of length $d_1 + 3$, obtained above, to obtain the missing cycles of length $d_1 + 2$ and $d_1 + 1$. Thus we may assume that there is at most one splice vertex; that is, C_1 has been grown at most once.

If C_1 has been grown exactly once, then there is only a problem with the $(d_1 + 1)$ -cycle, since again the splice vertex can be removed from the $(d_1 + 3)$ -cycle to give the $(d_1 + 2)$ -cycle. That is, the only cycles that might be missing have length at most $|C_1| + 2$. But these lengths are guaranteed by Lemma 3.

Case 2: $d_1 \geq \delta + 1$ and $k \geq 6$.

Unlike the first case, we may encounter several problematic cycles. We show in general how to proceed from a problematic cycle D_i to a longer cycle C_{i+1} , while attaining all cycle lengths in between. The cycle C_{i+1} is then grown as before until another problematic cycle D_{i+1} is encountered. Let $t_i = t(D_i)$, $d_i = |D_i|$ and $c_i = |C_i|$. Since $\delta \geq n + 3$, by Lemma 4, $d_i \leq n - 8$.

Case 2.1: $3 \leq t_i \leq \max(n - d_i - 4, 8)$.

Say the closest detourable pair is u and v with external neighbours u' and v' . Construct cycles by omitting from D_i the $t_i - 1$ internal vertices of the $u-v$ segment and adding paths of all lengths between u' and v' in $G - D_i$. One obtains all lengths from $d_i + 1$ up to $n - t_i + 1$.

If $t_i \leq 8$, let C_{i+1} be the cycle of length $n - t_i + 1$ so obtained. Otherwise, let C_{i+1} be the cycle of length $n - t_i - 3$ so obtained (so that cycles of length up to $c_{i+1} + 4$ have been found). Note that $c_{i+1} > d_i$. (In the latter case, $n - t_i - 3 \geq d_i + 4 - 3$.)

Case 2.2: $t_i \leq 2$.

Proceed as in Case 2.1. Then one obtains only all lengths from $d_i + 4 - t_i$ up to $n - t_i + 1$. If $i > 1$, then the missing cycles were obtained at the previous iteration (if $d_i \leq c_i + 2$), or can be constructed by deleting one or two splice vertices from the cycle of length $d_i + 4 - t_i$ (if $d_i \geq c_i + 4$), or a combination of the two (if $d_i = c_i + 3$). So assume $i = 1$.

Suppose $d_1 \leq 2k + 3$. Then one has the system of inequalities:

$$\begin{aligned} d_1 &\leq 2k + 3, \quad d_1 \geq \delta + 1, \quad \delta - (n - d_1 - 1) \leq d_1/2, \quad k \geq 6, \quad \text{and} \\ \delta &\geq (n - 7)/2 + k + \varepsilon_k. \end{aligned}$$

By linear programming, this system has no solution. So $d_1 \geq 2k + 4 \geq c_1 + 4$. (See
 So one obtains the missing cycles by deleting one or two splice vertices from the (Check B1)
 $(d_1 + 4 - t_1)$ -cycle.

Let C_{i+1} be the cycle of length $n - t_i + 1$ so obtained. Note that it has length
 at least $n - 1$ and is therefore not problematic.

Case 2.3: $t_i > \max(n - d_i - 4, 8)$ or t_i does not exist.

We show that this subcase cannot occur. Apply Lemma 5a with $T = 8$. It follows
 that

$$8d_i \leq 9n + 8k - 9\delta - 9. \quad (5)$$

Now, suppose $n - d_i \leq k + 1$. Then we have the system of inequalities:

$$\begin{aligned} n - d_i &\leq k + 1, \quad (5), \quad d_i \geq \delta + 1, \quad \delta - (n - d_i - 1) \leq d_i/2, \quad k \geq 6, \text{ and} \\ \delta &\geq (n - 7)/2 + k + \varepsilon_k. \end{aligned}$$

Linear programming shows that this system has no solution. So $n - d_i \geq k + 2$. (See
 That is, there are at least $k + 2$ external vertices. Note that every external vertex (Check B2)
 has at least $\delta - (n - d_i - 1)$ neighbours on the cycle, and $\delta - (n - d_i - 1) \geq$
 $2\delta + 2 - n \geq 2k - 5 + 2\varepsilon_k \geq k + 2$.

By Lemma 5b, it follows that t_i exists and $t_i \leq d_i/2 - 3$. In particular,
 $t_i \leq n - d_i - 4$, as a calculation shows; so this subcase cannot occur. (See

(Check B3)

Case 3: $d_1 \geq \delta + 1$ and $k = 5$.

This is similar to Case 2, but the details are messier. Define d_i , c_i and t_i as in
 Case 2. Note that now we have the bound $d_i \leq n - 5$, by Lemma 4.

Case 3.0: $d_i = n - 5$.

Since $\delta \geq (n + 3)/2$, every external vertex must be adjacent to every other
 external vertex and to exactly half of the vertices of D_i (in fact, every alternate
 vertex on D_i). In particular, there must be two external vertices x and y with
 the same set of neighbours on D_i . We claim that can find in this neighbourhood
 vertices u , u^{++} such that the intervening vertex u^+ is not in S . This follows
 trivially if $d_i \geq 11$. If $d_i \leq 10$ then, since $n - 5 = d_i \geq \delta + 1 \geq (n + 5)/2$, it
 follows that $n = 15$. And the exception is easily handled.

Then replace the uu^+u^{++} segment by the segment $uxyu^{++}$. The resultant
 cycle has length at least $n - 4$ and one cannot encounter another problematic
 cycle.

Case 3.1: $3 \leq t_i \leq \max(n - d_i - 4, 6)$.

Identical to Case 2.1, except at the end, the choice of C_{i+1} is based on whether $t_i \leq 6$ or not.

Case 3.2: $t_i = 2$.

Repeat the argument of Case 2.2. Again we obtain all cycles up to $n - 1$, except in the case that $i = 1$ when the cycles of $d_1 + 1$ and $d_1 + 2$ might be missing. If $d_1 \geq c_1 + 4$ then the missing cycles are obtained by deleting splice vertices from the $d_1 + 4 - t_i$ cycle. So we may assume that $d_1 \leq c_1 + 3$.

In particular, $d_1 \leq 13$. Then one has a system of inequalities:

$$\begin{aligned} d_1 &\leq 13, \quad d_1 \geq \delta + 1, \quad \delta - (n - d_1 - 1) \leq d_1/2, \quad d_1 \leq n - 6, \quad \text{and} \\ \delta &\geq (n + 3)/2. \end{aligned}$$

By linear programming, this system has no solution for $n \geq 22$. By brute force there are 6 integral solutions namely (n, d_1, δ) being $(17, 11, 10)$, $(18, 12, 11)$, $(19, 12, 11)$, $(19, 13, 11)$, $(20, 13, 12)$, $(21, 13, 12)$. (See Check C1)

Since $d_1 \leq c_1 + 3$, it follows that $c_1 \geq 8$. Thus at least 4 vertices of S have nonneighbours in S , and in the case where $d_1 = \delta + 1$ these 4 have external neighbours. In the case where $d_1 = \delta + 2$, then $d_1 = 13$ so that $c_1 = 10$. It follows that all vertices of S have two nonneighbours in S , and thus have external neighbours. In either case, there are at least 4 pairs of consecutive vertices of S which form detourable pairs. So we can choose a detourable pair such that the segment between them either misses the unique splice vertex (in which case the cycle of length $d_1 + 1 = c_1 + 2$ is provided by Lemma 3), or misses at least two splice vertices, or has length at least 3. In each case the missing cycles are obtained.

Case 3.3: t_i exists but $t_i > \max(n - d_i - 4, 6)$.

Suppose some external vertex has at most 6 neighbours on the cycle. It follows that $\delta - (n - d_i - 1) \leq 6$, so that $d_i \leq (n + 7)/2$. Since t_i exists, there is a detourable pair u, v . Let A denote the segment between them on D_i that does not contain a vertex of S . By assumption, $t_i > n - d_i - 4 \geq n/2 - 15/2$.

By the minimality of t_i , vertices on the segment A have no external neighbour, and so, since $d_i \leq \delta + 2$, are adjacent to all but one vertex on the cycle. Since $t \geq 7$, there is a splice vertex u on A . The vertex u is adjacent to two consecutive vertices on $D_i - A$ and can be moved there. So, if we assume t_i as small as possible, we get a contradiction. To be more accurate, we may assume that D_i

is a cycle with the smallest value of $t(D_i)$, taken over all cycles of length d_i that meet S in the requisite order and with at least $\min(d_i - c_i, 4)$ splice vertices, and thence obtain a contradiction.

So (we may assume) every external vertex has at least 7 neighbours on the cycle D_i . Also, by Lemma 5a with $T = 6$, $6d_i \leq 7n + 23 - 7\delta$. Consider an external vertex x . Since t_i exists but $t_i \geq 7$, there is a detourable pair on the cycle with a segment of at least 6 vertices between them. The vertex x cannot be adjacent to any vertex on that segment; it follows that x is adjacent to at most $(d_1 - 5)/2$ vertices on the cycle. On the other hand, it is adjacent to at least $\delta - (n - d_1 - 1)$ vertices on the cycle.

Suppose there are only 6 external vertices. One thus has the system of inequalities:

$$d = n - 6, \delta \geq (n + 3)/2, d_1 \geq \delta + 1, 6d_i \leq 7n + 23 - 7\delta, \text{ and } \delta - (n - d_1 - 1) \leq (d_1 - 5)/2.$$

By linear programming, this system has no solution. So there are at least 7 external vertices. (See Check C2)

Now, applying Lemma 5b, it follows that $t_1 \leq d_1/2 - 5/2$. By linear programming followed by brute force search it follows that $t_1 \leq n - d_1 - 4$ or $t_1 \leq 6$ except for the one case $n = 29, d_1 = 19, \delta = 16$ and $t_1 = 7$. (See Check C3)

But re-examine the proof of Lemma 5a with $T = 6$. If X denotes the number of segments, then we showed that

$$\delta - (n - d_1 - 1) \leq X \leq 4 + \lceil (d_1 - 4)/7 \rceil.$$

In this instance, this means that $X = 7$. Equality in the upper bound implies that the 5 vertices of S are consecutive on D_1 . But then by starting at the two outermost vertices of S and going in opposite directions, it is easy to obtain a division into 6 segments each containing at most 7 vertices and each with no internal vertex in S . This yields a contradiction. That is, this case cannot occur.

Case 3.4: t_i does not exist.

Since there are at least 6 external vertices, and each external vertex has at least 5 neighbours on the cycle, by the segmentation argument used previously it follows that each segment starting at a vertex of S has exactly one vertex adjacent to external vertices, and that vertex is adjacent to all external vertices. In particular, $d = \delta + 1 = (n + 5)/2$. Call these vertices the *gates*.

By the minimum degree condition, the remaining vertices on the cycle form a clique, and are adjacent to every gate. In particular, one can reorganise D_i such that the gates occur anywhere one chooses on the cycle, and so t_i exists and in fact can be chosen to be 4, so that we are done. QED

3.4 Sufficiency for k large

Lemma 7 *Let G be a graph with order n and minimum degree δ , and let k be an integer with $k \geq (n+6)/3$. If $\delta \geq (3n+k-9)/4$, then G is k -ordered-pancyclic, except when $k = n-3$ when $\delta \geq n-2$ is required.*

PROOF. (a) Let S be an ordered set of k vertices. We start by showing the existence of the smallest cycle through S .

Let $r = \delta - (k-1)$, $t = n - k$ and $u = \delta - (t-1)$. It will be useful to note that the lower bound on the minimum degree is equivalent to

$$k - u \leq (2r + 5 - t)/2. \quad (6)$$

Let $\mathcal{M} = \{M_1, \dots, M_m\}$ be the set of consecutive pairs of vertices of S that are not adjacent. If $m > n - k$ there is nothing to show, so assume $m \leq n - k$. Then we need to find m distinct vertices of $T = V(G) - S$ such that each is a common neighbour of a different pair of $\mathcal{M}$.

Let C_i be the set of joint neighbours in T of the two vertices of M_i . For $J \subseteq \{1, \dots, m\}$, let $C(J) = \bigcup_{i \in J} C_i$. By Hall's theorem, it is necessary and sufficient to show that

$$|C(J)| \geq |J| \text{ for all } J.$$

Fix any J , $|J| = j$. Let $\mathcal{M}_J$ denote the set $\{M_i : i \in J\}$; this may be viewed as the edge set of a graph with vertex set S . Since $|T| = n - k \geq m \geq j$, we may assume that $C(J) \neq T$.

Consider a vertex $x \in T - C(J)$. Then, for each $i \in J$, at least one of the vertices of M_i is nonadjacent to x . That is, $S - N(x)$ covers $\mathcal{M}_J$. In particular, x has at least $j/2$ nonneighbours in S . But x has at most $k - u$ nonneighbours in S , so that $j/2 \leq k - u$. By inequality (6) above, it follows that (we need only consider)

$$j \leq 2r + 5 - t.$$

(See
Check D1)

For each M_i , let z_i denote the number of other edges of $\mathcal{M}_J$ involving vertices of M_i . Note that $z_i \in \{0, 1, 2\}$. Then the degree-sum of M_i to T is at least $2r + 2 + z_i$. Now, let z_{max} denote the maximum z_i for $i \in J$. So

$$|C(J)| \geq 2r + 2 + z_{max} - t.$$

There are three cases.

Case 1: Assume $z_{max} = 2$.

Then, since $|C(J)| \geq 2r + 4 - t$, we may assume $j > 2r + 4 - t$. So we are done unless $j = 2r + 5 - t$. If $x \in T - C(J)$, then its nonneighbours, which number at most $k - u$, must cover $\mathcal{M}_J$ and so must number at least $j/2$. By inequality (6) above, it follows that $k - u = j/2$ and x has exactly $j/2$ nonneighbours. If $j = k$, then a short calculation yields a contradiction.

(See
Check D2)

So we may assume $j < k$. Thus the graph induced by $\mathcal{M}_J$ is acyclic. For there to be a cover of cardinality $j/2$, the set $\mathcal{M}_J$ must induce a union of even paths. But then there is a unique cover of cardinality $j/2$, call it D . In particular, there is no edge between $T - C(J)$ and D .

Let $v \in D$. Then because $\mathcal{M}_J$ is a union of even paths, v is incident with two edges in $\mathcal{M}_J$; that is, v has two nonneighbours in S . So v has at least $r + 2$ neighbours in T . By above, these neighbours are confined to $C(J)$, which has cardinality at most $j - 1$, so that $r + 2 \leq j - 1$. This yields the system of inequalities:

$$k \geq n/3 + 2, \delta \geq (3n + k - 9)/4, r + 2 \leq j - 1, \text{ and } j \leq n - k.$$

By linear programming this system has no solution, and so we have a contradiction.

(See
Check D3)

Case 2: Assume $z_{max} = 1$.

Then the set $\mathcal{M}_J$ induces a union of paths of length 1 or 2. Also we may assume that $j > 2r + 3 - t$. Now, if $x \in T - C(J)$, its nonneighbours, which number at most $k - u$, must cover $\mathcal{M}_J$. As $k - u \leq (j + 1)/2$, it follows that, if D is the set of centres of each path of length 2, all of D is in every minimum cover. In particular, there is no edge between $T - C(J)$ and D .

If $j = 1$, a calculation shows a contradiction, so D is nonempty. Then consider a vertex $v \in D$; one obtains almost the same system of inequalities as in Case 1. This yields a contradiction.

(See
Check D4)

Case 3: Assume $z_{\max} = 0$.

Then we may assume $j > 2r + 2 - t$. But $\mathcal{M}_J$ is the union of disjoint edges. So if $x \in T - C(J)$, then x has at least j nonneighbours, and so $k - u \geq j$. For the system of inequalities,

$$k \geq n/3 + 2, \delta \geq (3n + k - 9)/4, j \leq n - k, 2r + 3 - t \leq j \leq 2r + 5 - t, \\ \text{and } k - u \geq j,$$

a calculation yields a contradiction, except for $j = 2$ and $k = n - 3$ when we get the exceptional case mentioned in the hypothesis. (See

Check D5)

We have shown that $|C(J)| \geq |J|$ for all J . Thus, there is a cycle of length $k + m$ which meets the vertices of S in the required order.

(b) Now, we claim one can extend this cycle by repeatedly inserting an exterior vertex. For, suppose the current cycle has length d , and is problematic. Then we have the system of inequalities:

$$\delta - (n - d - 1) \leq d/2, 2(\delta - d + 1) \leq n - d, k \geq n/3 + 2, \text{ and } \\ \delta \geq (3n + k - 9)/4.$$

Linear programming shows this system has no solution for $n \geq 10$; and a short calculation shows that there is no integral solution for $n \leq 9$. This yields a contradiction. QED

(See
Check D6)

Theorem 3 is a consequence of Lemmas 6 and 7.

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